

Unit 109 - Field Data Review

Mohawk Valley West Slope demonstration. Method **Tractor**, 72.9 ac, BAF 20 variable-radius cruise on a 300 ft grid, 34 plots, 364 trees tallied. Simulated data with planted recording errors; see docs/methods.md.

Table 1. Stand summary from the BAF 20 cruise, per acre

STAND DENSITY		STAND INFO	
Basal area	214 sq ft/ac (SE 4.3 %)	Type	Sierran mixed conifer
Trees per acre	65	CWHR size / density	5 / D (cover 88 %)
QMD	24.6 in	SDI	263 (44 % of max 592)
SAWTIMBER (>= 10 in)		BIOMASS (< 10 in)	
BA QMD	213 sq ft/ac 25.1 in	BA QMD	1 sq ft/ac 8.8 in
TPA	62	TPA	3
Volume	51.1 MBF/ac (9290 cu ft)	Volume	0.9 green tons/ac (29 cu ft)
TREATMENT TARGET		CRUISE DESIGN	
Leave	35 % of max SDI = ~169 sq ft/ac BA	Sampling error	8.8 % at 95 % (t = 2.03, n = 34)
Species (BA)	WF 70, PP 55, DF 36, IC 32, SP 21	Stratum standard	40 % - MEETS; 34 plots vs 20-plot practice: ok

Cubic volume by form factor 0.42 net of defect; Scribner board feet at 5.5 bf/cu ft (Keegan et al. 2010); green tons at 60 lb/cu ft. SDI in the summation form; maximum is the basal-area-weighted FVS Western Sierra species maximum. Sampling error at 95 % confidence with Student's t; stratum standard 40 % for tree-measurement sales (FSH 2409.12 ch. 40, 41.1).

Table 2. Stand table by 4-inch DBH class, per acre

DBH class, in	Trees	TPA	BA	cu ft	MBF
8-11.9	4	4.3	2.4	64	0.4
12-15.9	17	9.4	10.0	292	1.6
16-19.9	40	12.7	23.5	807	4.4
20-23.9	68	15.6	40.0	1506	8.3
24-27.9	44	7.1	25.9	1025	5.6
28-31.9	49	6.0	28.8	1271	7.0
32-35.9	39	3.7	22.9	1087	6.0
36-39.9	31	2.3	18.2	922	5.1
40-43.9	25	1.5	14.7	767	4.2
44-47.9	13	0.7	7.6	408	2.2
48-51.9	9	0.4	5.3	293	1.6
52-55.9	9	0.3	5.3	297	1.6
56-59.9	6	0.2	3.5	210	1.2
60-63.9	10	0.3	5.9	371	2.0
All	364	64.7	214.1	9319	51.3

Table 3. Stock table by species, per acre

Species	Trees	TPA	BA	QMD	cu ft	MBF
WF	119	21.1	70.0	24.7	3037	16.7
PP	94	16.0	55.3	25.2	2393	13.2
DF	62	11.2	36.5	24.4	1576	8.7
IC	54	10.5	31.8	23.5	1377	7.6
SP	35	5.9	20.6	25.3	935	5.1
All	364	64.7	214.1	24.6	9319	51.3

Trees = trees tallied in the cruise; TPA, BA (sq ft/ac), cubic feet and Scribner MBF per acre by BAF expansion. Species: PP ponderosa pine, WF white fir, DF Douglas-fir, SP sugar pine, IC incense-cedar.

Unit 109: plots on the 300 ft grid (orange = QA flag)

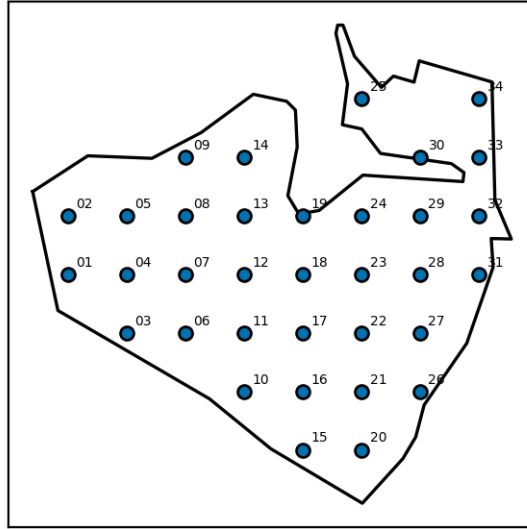


Figure 1. Plot locations with QA status.

QA findings: no issues found; unit accepted

Table 4. Plot list

Plot	Trees in	BA sq ft/ac	Slope %	Status
109-01	15	300	27	ok
109-02	10	200	42	ok
109-03	10	200	102	ok
109-04	11	220	16	ok
109-05	9	180	22	ok
109-06	12	240	104	ok
109-07	3	60	18	ok
109-08	12	240	21	ok
109-09	12	240	52	ok
109-10	11	220	69	ok
109-11	13	260	55	ok
109-12	11	220	31	ok
109-13	15	300	16	ok
109-14	13	260	25	ok
109-15	10	200	62	ok
109-16	10	200	32	ok
109-17	10	200	47	ok
109-18	10	200	24	ok
109-19	7	140	229	ok
109-20	12	240	39	ok
109-21	12	240	25	ok
109-22	9	180	8	ok
109-23	11	220	17	ok
109-24	13	260	86	ok
109-25	10	200	63	ok
109-26	10	200	75	ok
109-27	10	200	52	ok
109-28	14	280	42	ok
109-29	11	220	59	ok
109-30	11	220	81	ok
109-31	13	260	147	ok
109-32	2	40	185	ok
109-33	12	240	111	ok
109-34	10	200	31	ok